

Profile

Software Engineer with a systems engineering background and experience developing reliable applications, data pipelines, APIs, and internal tools. Proficient in Python, C#, and SQL, with experience designing and supporting software used in complex, production-critical manufacturing environments. Strong problem solver who excels at understanding complex systems, designing maintainable solutions, and translating operational requirements into scalable software.

Professional Experience

System Engineer 2, AESC US — Smyrna, TN
May 2025 — Present

- Develop production software, MES functionality, data pipelines, automation, and internal applications using **Python, C#, .NET, React, and SQL Server**.
- Designed and developed a **Python-based ETL pipeline** to ingest, transform, and consolidate millions of machine-generated production records into SQL Server for manufacturing analysis and KPI tracking.
- Designed and developed full-stack applications using **.NET, React, and SQL Server** to automate workflows, improve data visibility, and standardize operational processes.
- Built a **Python automation framework** with structured logging, automated email alerting, reporting, and file processing, replacing manual operational workflows.
- Own the reliability and continuous improvement of production systems, diagnosing issues across applications, databases, integrations, and infrastructure while collaborating with engineering, operations, QA, and vendors.

System Engineer 1, AESC US — Smyrna, TN
May 2024 — May 2025

- Supported the installation, configuration, and production startup of a new **DELMIA Apriso MES platform**, troubleshooting software and system issues during early production operations.
- Developed custom **DELMIA Apriso functionality** to automate manufacturing workflows and improve data tracking and visibility for MFG and QA teams.
- Developed internal tools and data-processing workflows using **Python and SQL** to reduce manual processes and improve operational efficiency.
- Investigated production issues across applications, databases, and workflows, identifying root causes and coordinating solutions with internal teams and software vendors.

System Engineering Intern, AESC US — Smyrna, TN
Feb 2023 — May 2024

- Provided technical support for production systems, including legacy MES troubleshooting, IPC restoration, and workstation support.
- Executed SQL-based data updates under supervision and gained hands-on experience with MES environments, manufacturing systems, and production workflows.

Education

Middle Tennessee State University — Murfreesboro, TN
B.S. Computer Science, Aug 2021 — Dec 2024

Motlow State Community College — Smyrna, TN
A.A.S. Mechatronics, Aug 2020 — May 2021

Technical Skills

Languages: Python, C#, SQL, C/C++
Backend & Web: .NET, ASP.NET Core, Entity Framework Core, React, REST APIs
Data: Microsoft SQL Server, ETL, Data Processing, Data Transformation
Engineering: Full-Stack Development, Automation, Internal Tooling, System Integration, Root-Cause Analysis
Manufacturing: MES, DELMIA Apriso, Production Systems
Tools: Git, IIS, Windows Server, Rocky Linux/Red Hat Enterprise Linux, Grafana